

Manual Change History (4022102) QSK38, QSK50, and QSK60 CM2150 Electronic Control System Troubleshooting and Repair Manual

Date	17-Aug-2010
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-474 (/qs3/pubsys2/xml/en/procedures/99/99-019-474.html)
Procedure Title	Bosch Aftertreatment DEF Controller Connector
Update Reason	Added procedure to outline.

Date	28-Jun-2010
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc2998 (/qs3/pubsys2/xml/en/procedures/122/122-fc2998.html)
Procedure Title	Engine Torque Limit Feature - Special Instructions
Update Reason	Added fault code to outline.

Date	28-Jun-2010
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc3696 (/qs3/pubsys2/xml/en/procedures/122/122-fc3696.html)
Procedure Title	Droop Adjustment Potentiometer Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	Added fault code to outline.

Date	21-Sep-2010
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc2619 (/qs3/pubsys2/xml/en/procedures/122/122-fc2619.html)
Procedure Title	Auxiliary Equipment Sensor Input 1 Engine Protection Critical - Special Instructions
Update Reason	Added fault code to outline.
Date	21-Sep-2010
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-2619 (/qs3/pubsys2/xml/en/procedures/122/122-t05-2619.html)
Procedure Title	Auxiliary Equipment Sensor Input 1 Engine Protection Critical - Special Instructions
Update Reason	Added fault code to outline.
Date	21-Sep-2010
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1543 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1543.html)
Procedure Title	Auxiliary Pressure Sensor Input 1 Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated procedure by adding an additional step to check the resistive jumper.
Date	13-Dec-2010
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc1774 (/qs3/pubsys2/xml/en/procedures/122/122-fc1774.html)
Procedure Title	Oil Priming Pump Control Circuit - Voltage Above Normal or Shorted to High Source
Update Reason	New FC1774 added to the Troubleshooting Section.

Date	13-Dec-2010
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1774 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1774.html)
Procedure Title	Oil Priming Pump Control Circuit - Voltage Above Normal or Shorted to High Source
Update Reason	New FC1774 added to the Troubleshooting Section.

Date	21-Jan-2011
Section	Section F - Familiarization
Procedure Number	00-022-999 (/qs3/pubsys2/xml/en/procedures/00/00-022-999.html)
Procedure Title	Service Tools and Hardware - Overview
Update Reason	Updated General Information, Initial Check, Setup, and Resistance Check sections.

Date	01-Apr-2011
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-235 (/qs3/pubsys2/xml/en/procedures/122/122-t05-235.html)
Procedure Title	Coolant Level - Data Valid But Below Normal Operating Range - Most Severe Level
Update Reason	Step 2B was added to the troubleshooting steps.

Date	02-Jun-2011
Section	Section TT -Troubleshooting Symptoms (New Format)
Procedure Number	122-t02-1001 (/qs3/pubsys2/xml/en/procedures/122/122-t02-1001.html)
Procedure Title	Engine Performance Troubleshooting Tree
Update Reason	Multiple step updates.

Date	12-Oct-2011
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc1852 (/qs3/pubsys2/xml/en/procedures/122/122-fc1852.html)
Procedure Title	Water in Fuel Indicator Sensor Circuit - Voltage Above Normal or Shorted to High Source
Update Reason	Updated Shop Talk
Date	08-Dec-2011
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	122-019-115 (/qs3/pubsys2/xml/en/procedures/122/122-019-115.html)
Procedure Title	Rail Fuel Pressure Sensor
Update Reason	Corrected the torque value for the fuel pump pressure sensor.
Date	17-Jan-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc1544 (/qs3/pubsys2/xml/en/procedures/122/122-fc1544.html)
Procedure Title	Auxiliary Pressure Sensor Input 1 - Special Instructions
Update Reason	Updated Shop Talk and circuit descriptions to include Marine specific information.
Date	13-Feb-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-2727 (/qs3/pubsys2/xml/en/procedures/122/122-t05-2727.html)
Procedure Title	SAE J1939 Engine Network - Abnormal Update Rate
Update Reason	Added New Step 2 to check the 4-pin power harness for both unswitched and keyswitched power.

Date	10-Feb-2012
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	105-019-032 (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)
Procedure Title	ECM Calibration Code
Update Reason	Updated to include information concerning troubleshooting of ECM calibration issues once new calibration is installed.
Date	19-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-559 (/qs3/pubsys2/xml/en/procedures/122/122-t05-559.html)
Procedure Title	Injector Metering Rail pressure - Data Valid But Below Normal Operating Range - Moderately Severe Level
Update Reason	Updated FCT by removing mechanical dump valve.
Date	10-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1911 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1911.html)
Procedure Title	Injector Metering Rail pressure - Data Valid But Below Normal Operating Range - Most Severe Level
Update Reason	Updated FCT by removing mechanical dump valve.
Date	01-Jun-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-245 (/qs3/pubsys2/xml/en/procedures/122/122-t05-245.html)
Procedure Title	Fan Control Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	Changed Step 2B Specifications: Within the OEM resistance specification?

Date	23-Oct-2012
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	122-t02-1001 (/qs3/pubsys2/xml/en/procedures/122/122-t02-1001.html)
Procedure Title	Engine Performance Troubleshooting Tree
Update Reason	Corrected "Next Step" numbering
Date	03-Dec-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-2261 (/qs3/pubsys2/xml/en/procedures/122/122-t05-2261.html)
Procedure Title	Fuel Pump Delivery Pressure - Data Valid But Above Normal Operating Range - Least Severe Level
Update Reason	Revised entire fault code.
Date	26-Feb-2013
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	122-019-042 (/qs3/pubsys2/xml/en/procedures/122/122-019-042.html)
Procedure Title	Engine Speed Sensor (ESS)
Update Reason	Added procedure to manual.
Date	25-Apr-2013
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-2215 (/qs3/pubsys2/xml/en/procedures/122/122-t05-2215.html)
Procedure Title	Fuel Pump Delivery Pressure - Data Valid But Below Operating Range - Moderately Severe Level
Update Reason	Replaced Guided Steps 2C and 2D and added Guided Step 2E.

Date	23-Apr-2013
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-2262 (/qs3/pubsys2/xml/en/procedures/122/122-t05-2262.html)
Procedure Title	Fuel Pump Delivery Pressure - Data Valid But Below Operating Range - Least Severe Level
Update Reason	Replaced Guided Steps 2C and 2D and added Guided Step 2E.

Date	28-Jun-2013
Section	Section 19 - Electronic Controls
Procedure Number	122-022-001_19 (/qs3/pubsys2/xml/en/procedures/122/122-022-001_19.html)
Procedure Title	Service Tools
Update Reason	Changed tool description to INSITE™ Electronic Service Tool Software Kit (3886388).

Date	28-Jun-2013
Section	Section 19 - Electronic Controls
Procedure Number	122-019-427 (/qs3/pubsys2/xml/en/procedures/122/122-019-427.html)
Procedure Title	ECM ROM Boot
Update Reason	Revise table: Change CM850 to CM2150, Part Number 4918583.

Date	05-Aug-2013
Section	Section TT - Troubleshooting Performance Troubleshooting Tree - New Format)
Procedure Number	122-t02-1001 (/qs3/pubsys2/xml/en/procedures/122/122-t02-1001.html)
Procedure Title	Engine Performance Troubleshooting Tree - CM2150 Electronic Control System
Update Reason	Guided Step 3H: Added, Repair fuel rail or leaking component to YES.

Date	15-May-2014
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1521thermistor (/qs3/pubsys2/xml/en/procedures/122/122-t05-1521thermistor.html)
Procedure Title	FAULT CODE 1521 (Thermistor) Exhaust Gas Temperature Sensor Circuit Cylinder 3 - Voltage Above Normal or Shorted to High Source
Update Reason	Added steps 3E and 3F.
Date	15-May-2014
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1522thermistor (/qs3/pubsys2/xml/en/procedures/122/122-t05-1522thermistor.html)
Procedure Title	FAULT CODE 1522 (Thermistor) Exhaust Gas Temperature Sensor Circuit Cylinder 5 - Voltage Above Normal or Shorted to High Source
Update Reason	Added steps 3E and 3F.
Date	15-May-2014
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1523thermistor (/qs3/pubsys2/xml/en/procedures/122/122-t05-1523thermistor.html)
Procedure Title	FAULT CODE 1523 (Thermistor) Exhaust Gas Temperature Sensor Circuit Cylinder 7 - Voltage Above Normal or Shorted to High Source
Update Reason	Added steps 3E and 3F.

Date	15-May-2014
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1524thermistor (/qs3/pubsys2/xml/en/procedures/122/122-t05-1524thermistor.html)
Procedure Title	FAULT CODE 1524 (Thermistor) Exhaust Gas Temperature Sensor Circuit Cylinder 9 - Voltage Above Normal or Shorted to High Source
Update Reason	Added steps 3E and 3F.
Date	15-May-2014
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1525thermistor (/qs3/pubsys2/xml/en/procedures/122/122-t05-1525thermistor.html)
Procedure Title	FAULT CODE 1525 (Thermistor) Exhaust Gas Temperature Sensor Circuit Cylinder 11 - Voltage Above Normal or Shorted to High Source
Update Reason	Added steps 3E and 3F.
Date	15-May-2014
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1526thermistor (/qs3/pubsys2/xml/en/procedures/122/122-t05-1526thermistor.html)
Procedure Title	FAULT CODE 1526 (Thermistor) Exhaust Gas Temperature Sensor Circuit Cylinder 13 - Voltage Above Normal or Shorted to High Source
Update Reason	Added steps 3E and 3F.

Date	15-May-2014
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1527thermistor (/qs3/pubsys2/xml/en/procedures/122/122-t05-1527thermistor.html)
Procedure Title	FAULT CODE 1527 (Thermistor) Exhaust Gas Temperature Sensor Circuit Cylinder 15 - Voltage Above Normal or Shorted to High Source
Update Reason	Added steps 3E and 3F.
Date	15-May-2014
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1529thermistor (/qs3/pubsys2/xml/en/procedures/122/122-t05-1529thermistor.html)
Procedure Title	FAULT CODE 1529 (Thermistor) Exhaust Gas Temperature Sensor Circuit Cylinder 2 - Voltage Above Normal or Shorted to High Source
Update Reason	Added steps 3E and 3F.
Date	14-Apr-2015
Section	Section F - Familiarization
Procedure Number	122-101-007 (/qs3/pubsys2/xml/en/procedures/122/122-101-007.html)
Procedure Title	Electronic Controlled Fuel System
Update Reason	Replaced all CM500 fuel system references to QSK fuel system as this is generic description.
Date	04-Mar-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc4615 (/qs3/pubsys2/xml/en/procedures/122/122-fc4615.html)
Procedure Title	Fuel Pump Delivery Pressure - Data Valid But Above Normal Operating Range - Most Severe Level
Update Reason	Added to outline.

Date	04-Mar-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-4615 (/qs3/pubsys2/xml/en/procedures/122/122-t05-4615.html)
Procedure Title	Fuel Pump Delivery Pressure - Data Valid But Above Normal Operating Range - Most Severe Level
Update Reason	Added to outline.
Date	10-Apr-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc1911 (/qs3/pubsys2/xml/en/procedures/122/122-fc1911.html)
Procedure Title	Injector Metering Rail 1 Pressure - Data Valid but Above Normal Operating Range - Most Severe Level
Update Reason	Updated to include test procedure for electronically actuated injectors.
Date	07-Apr-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc449 (/qs3/pubsys2/xml/en/procedures/122/122-fc449.html)
Procedure Title	Injector Metering Rail 1 Pressure - Data Valid but Above Normal Operating Range - Most Severe Level
Update Reason	Updated to include test procedure for electronically actuated injectors.
Date	07-Apr-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc559 (/qs3/pubsys2/xml/en/procedures/122/122-fc559.html)
Procedure Title	Injector Metering Rail 1 Pressure - Data Valid but Below Normal Operating Range - Moderately Severe Level
Update Reason	Updated to include test procedure for electronically actuated injectors.

Date	07-Apr-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc2215 (/qs3/pubsys2/xml/en/procedures/122/122-fc2215.html)
Procedure Title	Fuel Pump Delivery Pressure - Data Valid but Below Operating Range - Moderately Severe Level
Update Reason	Updated to include test procedure for electronically actuated injectors.

Date	07-Apr-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc2262 (/qs3/pubsys2/xml/en/procedures/122/122-fc2262.html)
Procedure Title	Fuel Pump Delivery Pressure - Data Valid but Below Operating Range - Least Severe Level
Update Reason	Updated to include test procedure for electronically actuated injectors.

Date	25-Jun-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc419 (/qs3/pubsys2/xml/en/procedures/122/122-fc419.html)
Procedure Title	Intake Manifold Pressure Bank Imbalance - Data Erratic, Intermittent, or Incorrect
Update Reason	Added to outline.

Date	25-Jun-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-419 (/qs3/pubsys2/xml/en/procedures/122/122-t05-419.html)
Procedure Title	Intake Manifold Pressure Bank Imbalance - Data Erratic, Intermittent, or Incorrect
Update Reason	Added to outline.

Date	25-Jun-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc3722 (/qs3/pubsys2/xml/en/procedures/122/122-fc3722.html)
Procedure Title	Intake Manifold Pressure Bank Imbalance - Data Erratic, Intermittent, or Incorrect
Update Reason	Added to outline.
Date	25-Jun-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-3722 (/qs3/pubsys2/xml/en/procedures/122/122-t05-3722.html)
Procedure Title	Intake Manifold Pressure Bank Imbalance - Data Erratic, Intermittent, or Incorrect
Update Reason	Added to outline.
Date	03-Sep-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc161 (/qs3/pubsys2/xml/en/procedures/122/122-fc161.html)
Procedure Title	Intake Manifold 3 Temperature Sensor Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated the Component Location from 'front' to 'rear'.
Date	18-Sep-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-2727 (/qs3/pubsys2/xml/en/procedures/122/122-t05-2727.html)
Procedure Title	SAE J1939 Data Link 2 Engine Network - Abnormal Update Rate
Update Reason	Updated guided steps 2A, 2B, 2C, 2D, 2E, and 2F.

Date	24-Sep-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc1852 (/qs3/pubsys2/xml/en/procedures/122/122-fc1852.html)
Procedure Title	Water In Fuel Indicator - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Added information on engine derate for marine engines.

Date	24-Sep-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	00-t05-1852 (/qs3/pubsys2/xml/en/procedures/00/00-t05-1852.html)
Procedure Title	Water In Fuel Indicator - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Replaced 122-t05-1852.

Date	24-Sep-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc4642 (/qs3/pubsys2/xml/en/procedures/122/122-fc4642.html)
Procedure Title	Water in Fuel Indicator - Data Valid But Above Normal Operating Range - Most Severe Level
Update Reason	Added fault code to manual.

Date	06-Oct-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5634 (/qs3/pubsys2/xml/en/procedures/122/122-fc5634.html)
Procedure Title	Marine Overload Protection - Condition Exists - Choice
Update Reason	Added to manual.

Date	06-Oct-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5634aux (/qs3/pubsys2/xml/en/procedures/122/122-fc5634aux.html)
Procedure Title	Marine Overload Protection - Condition Exists
Update Reason	Added to manual.

Date	05-Oct-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5634prop (/qs3/pubsys2/xml/en/procedures/122/122-fc5634prop.html)
Procedure Title	Marine Overload Protection - Condition Exists
Update Reason	Added to manual.

Date	06-Oct-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5635 (/qs3/pubsys2/xml/en/procedures/122/122-fc5635.html)
Procedure Title	Marine Overload Protection - Special Instructions - Choice
Update Reason	Added to manual.

Date	30-Sep-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5635aux (/qs3/pubsys2/xml/en/procedures/122/122-fc5635aux.html)
Procedure Title	Marine Overload Protection - Special Instructions
Update Reason	Added to manual.

Date	09-Sep-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5635prop (/qs3/pubsys2/xml/en/procedures/122/122-fc5635prop.html)
Procedure Title	Marine Overload Protection - Special Instructions
Update Reason	Added to manual.
Date	17-Nov-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc418 (/qs3/pubsys2/xml/en/procedures/122/122-fc418.html)
Procedure Title	Water in Fuel Indicator - Data Valid But Above Normal Operating Range - Least Severe Level
Update Reason	Added to manual.
Date	17-Nov-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t418 (/qs3/pubsys2/xml/en/procedures/122/122-t418.html)
Procedure Title	Water in Fuel Indicator - Data Valid But Above Normal Operating Range - Least Severe Level
Update Reason	Added to manual.
Date	28-Jul-2016
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-111 (/qs3/pubsys2/xml/en/procedures/122/122-t05-111.html)
Procedure Title	Engine Control Module Critical Internal Failure - Bad Intelligent Device or Component
Update Reason	Added to manual.

Date	08-Aug-2016
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc111 (/qs3/pubsys2/xml/en/procedures/122/122-fc111.html)
Procedure Title	Engine Control Module Critical Internal Failure - Bad Intelligent Device or Component
Update Reason	Added to manual.
Date	26-Aug-2016
Section	Section 19 - Electronic Controls
Procedure Number	122-019-013 (/qs3/pubsys2/xml/en/procedures/122/122-019-013.html)
Procedure Title	Exhaust Temperature Sensor
Update Reason	Change Torque Value for exhaust temperature sensor and wire clamp capscrews.
Date	31-Oct-2016
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	122-019-043 (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)
Procedure Title	Engine Wiring Harness
Update Reason	Updated for new harness bracket design. Add Installation Instruction, Bulletin 5414606.
Date	3-Nov-2016
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5633 (/qs3/pubsys2/xml/en/procedures/122/122-fc5633.html)
Procedure Title	Marine Low Load Protection - Condition Exists - Choice
Update Reason	Added to fault code to outline.

Date	3-Nov-2016
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5633aux (/qs3/pubsys2/xml/en/procedures/122/122-fc5633aux.html)
Procedure Title	Marine Low Load Protection - Condition Exists
Update Reason	Added to fault code to outline.
Date	3-Nov-2016
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5633prop (/qs3/pubsys2/xml/en/procedures/122/122-fc5633prop.html)
Procedure Title	Marine Low Load Protection - Condition Exists
Update Reason	Added to fault code to outline.
Date	3-Nov-2016
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	00-t05-5633 (/qs3/pubsys2/xml/en/procedures/00/00-t05-5633.html)
Procedure Title	Marine Low Load Protection - Condition Exists
Update Reason	Added to fault code to outline.
Date	05-Dec-2016
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	00-ttecm (/qs3/pubsys2/xml/en/procedures/00/00-ttecm.html)
Procedure Title	ECM - No Communication Troubleshooting Tree
Update Reason	Modified Steps 2, 6, 9 and 11 and added information for new INLINE™ 7 data link.

Date	13-Dec-2016
Section	Section F - Familiarization
Procedure Number	00-022-999 (/qs3/pubsys2/xml/en/procedures/00/00-022-999.html)
Procedure Title	Service Tools and Hardware - Overview
Update Reason	Removed or corrected information for INLINE™ 4, 5, and 6. Added new information for INLINE™ 7 data link.
Date	10-Jan-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc521 (/qs3/pubsys2/xml/en/procedures/122/122-fc521.html)
Procedure Title	Transmission Oil Temperature Sensor Circuit - Voltage Above Normal or Shorted to High Source
Update Reason	Added fault code to manual.
Date	10-Jan-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc522 (/qs3/pubsys2/xml/en/procedures/122/122-fc522.html)
Procedure Title	Transmission Oil Temperature Sensor Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	Added fault code to manual.
Date	10-Jan-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc2562 (/qs3/pubsys2/xml/en/procedures/122/122-fc2562.html)
Procedure Title	Transmission (or Gearbox) Oil Temperature - Data Valid But Above Normal Operating Range - Most Severe Level
Update Reason	Added fault code to manual.

Date	10-Jan-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc4688 (/qs3/pubsys2/xml/en/procedures/122/122-fc4688.html)
Procedure Title	Water in Fuel Indicator 2 Sensor Circuit - Voltage Above Normal or Shorted to High Source
Update Reason	Added fault code to manual.

Date	10-Jan-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc4689 (/qs3/pubsys2/xml/en/procedures/122/122-fc4689.html)
Procedure Title	Water in Fuel Indicator 2 Sensor Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	Added fault code to manual.

Date	10-Jan-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5366 (/qs3/pubsys2/xml/en/procedures/122/122-fc5366.html)
Procedure Title	Water in Fuel Indicator 2 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Added fault code to manual.

Date	10-Jan-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5367 (/qs3/pubsys2/xml/en/procedures/122/122-fc5367.html)
Procedure Title	Water in Fuel Indicator 2 - Data Valid But Above Normal Operating Range - Most Severe Level
Update Reason	Added fault code to manual.

Date	10-Jan-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5932 (/qs3/pubsys2/xml/en/procedures/122/122-fc5932.html)
Procedure Title	Water in Fuel Indicator 2 - Data Valid But Above Normal Operating Range - Least Severe Level
Update Reason	Added fault code to manual.

Date	10-Jan-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5946 (/qs3/pubsys2/xml/en/procedures/122/122-fc5946.html)
Procedure Title	Transmission (or Gearbox) Oil Temperature - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Added fault code to manual.

Date	23-Jan-2017
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	122-019-657 (/qs3/pubsys2/xml/en/procedures/122/122-019-657.html)
Procedure Title	Exhaust Gas Temperature Sensor(s) Wiring Harness
Update Reason	Added to outline.

Date	23-Mar-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-2186ind (/qs3/pubsys2/xml/en/procedures/122/122-t05-2186ind.html)
Procedure Title	Sensor Supply 4 - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated sensor supply 4 circuit.

Date	23-Mar-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-2186pg (/qs3/pubsys2/xml/en/procedures/122/122-t05-2186pg.html)
Procedure Title	Sensor Supply 4 - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated sensor supply 4 circuit.
Date	21-Apr-2017
Section	Section 19 - Electronic Controls
Procedure Number	122-019-067 (/qs3/pubsys2/xml/en/procedures/122/122-019-067.html)
Procedure Title	Engine Oil Temperature Sensor
Update Reason	Added procedure to manual.
Date	24-Apr-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc6719 (/qs3/pubsys2/xml/en/procedures/122/122-fc6719.html)
Procedure Title	Fuel Pump Delivery Pressure - Data Valid But Above Normal Operating Range - Most Severe Level
Update Reason	Added to manual.
Date	24-Apr-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-6719 (/qs3/pubsys2/xml/en/procedures/122/122-t05-6719.html)
Procedure Title	Fuel Pump Delivery Pressure - Data Valid But Above Normal Operating Range - Most Severe Level
Update Reason	Added to manual.

Date	02-May-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc2216 (/qs3/pubsys2/xml/en/procedures/122/122-fc2216.html)
Procedure Title	Fuel Pump Delivery Pressure - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Added to manual.
Date	07-Aug-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-245 (/qs3/pubsys2/xml/en/procedures/122/122-t05-245.html)
Procedure Title	Fan Control Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	Revised Step 1A Specifications: Fault Code 245 active or inactive with more than one count in the last 25 engine hours?
Date	08-Sep-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-237 (/qs3/pubsys2/xml/en/procedures/122/122-t05-237.html)
Procedure Title	External Speed Command Input (Multiple Unit Synchronization) - Data Erratic, Intermittent, or Incorrect
Update Reason	Reworded Step 1B (NO) answer to reflect the correct direction per controls.
Date	09-Sep-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc595 (/qs3/pubsys2/xml/en/procedures/122/122-fc595.html)
Procedure Title	Turbocharger 1 Speed - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Added this fault code to the manual.

Date	15-Nov-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc1542 (/qs3/pubsys2/xml/en/procedures/122/122-fc1542.html)
Procedure Title	Auxiliary Pressure Sensor Input 1 Circuit - Voltage Above Normal or Shorted to High Source
Update Reason	Revised fault code to modify circuit art and add OBD information to the Overview. Also completely revised the troubleshooting tree portion of the fault code.
Date	10-May-2018
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc1362 (/qs3/pubsys2/xml/en/procedures/122/122-fc1362.html)
Procedure Title	Engine Oil Filter Differential Pressure - Data Valid but Above Normal Operating Range - Moderately Severe Level
Update Reason	Revised circuit art for QSK60 Industrial engine to match labels to wiring diagram.
Date	22-May-2018
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	122-019-365 (/qs3/pubsys2/xml/en/procedures/122/122-019-365.html)
Procedure Title	Crankshaft Position Sensor
Update Reason	Added to outline.
Date	27-Jun-2018
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-238sa00 (/qs3/pubsys2/xml/en/procedures/122/122-t05-238sa00.html)
Procedure Title	Sensor Supply 3 Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	Added Step 2B.

Date	27-Jun-2018
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-238sa10 (/qs3/pubsys2/xml/en/procedures/122/122-t05-238sa10.html)
Procedure Title	Sensor Supply 3 Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	Added Step 2B.
Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-322qsk38 (/qs3/pubsys2/xml/en/procedures/122/122-t05-322qsk38.html)
Procedure Title	Injector Solenoid Driver Cylinder 1 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.
Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-322qsk50-60 (/qs3/pubsys2/xml/en/procedures/122/122-t05-322qsk50-60.html)
Procedure Title	Injector Solenoid Driver Cylinder 1 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.
Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-323qsk38 (/qs3/pubsys2/xml/en/procedures/122/122-t05-323qsk38.html)
Procedure Title	Injector Solenoid Driver Cylinder 5 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-323qsk50-60 (/qs3/pubsys2/xml/en/procedures/122/122-t05-323qsk50-60.html)
Procedure Title	Injector Solenoid Driver Cylinder 5 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-324qsk38 (/qs3/pubsys2/xml/en/procedures/122/122-t05-324qsk38.html)
Procedure Title	Injector Solenoid Driver Cylinder 5 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-324qsk50-60 (/qs3/pubsys2/xml/en/procedures/122/122-t05-324qsk50-60.html)
Procedure Title	Injector Solenoid Driver Cylinder 3 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-325qsk38 (/qs3/pubsys2/xml/en/procedures/122/122-t05-325qsk38.html)
Procedure Title	Injector Solenoid Driver Cylinder 6 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-325qsk50-60 (/qs3/pubsys2/xml/en/procedures/122/122-t05-325qsk50-60.html)
Procedure Title	Injector Solenoid Driver Cylinder 6 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-331qsk38 (/qs3/pubsys2/xml/en/procedures/122/122-t05-331qsk38.html)
Procedure Title	Injector Solenoid Driver Cylinder 2 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-331qsk50-60 (/qs3/pubsys2/xml/en/procedures/122/122-t05-331qsk50-60.html)
Procedure Title	Injector Solenoid Driver Cylinder 2 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-332qsk38 (/qs3/pubsys2/xml/en/procedures/122/122-t05-332qsk38.html)
Procedure Title	Injector Solenoid Driver Cylinder 4 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-332qsk50-60 (/qs3/pubsys2/xml/en/procedures/122/122-t05-332qsk50-60.html)
Procedure Title	Injector Solenoid Driver Cylinder 4 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1548qsk38 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1548qsk38.html)
Procedure Title	Injector Solenoid Driver Cylinder 7 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1548qsk50-60 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1548qsk50-60.html)
Procedure Title	Injector Solenoid Driver Cylinder 7 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1549qsk38 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1549qsk38.html)
Procedure Title	Injector Solenoid Driver Cylinder 8 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1549qsk50-60 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1549qsk50-60.html)
Procedure Title	Injector Solenoid Driver Cylinder 8 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1551qsk38 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1551qsk38.html)
Procedure Title	Injector Solenoid Driver Cylinder 10 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1551qsk50-60 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1551qsk50-60.html)
Procedure Title	Injector Solenoid Driver Cylinder 10 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1552qsk38 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1552qsk38.html)
Procedure Title	Injector Solenoid Driver Cylinder 11 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1552qsk50-60 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1552qsk50-60.html)
Procedure Title	Injector Solenoid Driver Cylinder 11 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1553qsk38 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1553qsk38.html)
Procedure Title	Injector Solenoid Driver Cylinder 12 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1553qsk50-60 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1553qsk50-60.html)
Procedure Title	Injector Solenoid Driver Cylinder 12 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1554 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1554.html)
Procedure Title	Injector Solenoid Driver Cylinder 13 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1555 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1555.html)
Procedure Title	Injector Solenoid Driver Cylinder 14 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	04-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1556 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1556.html)
Procedure Title	Injector Solenoid Driver Cylinder 15 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	31-May-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1622qsk38 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1622qsk38.html)
Procedure Title	Injector Solenoid Driver Cylinder 16 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	31-May-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-1622qsk50-60 (/qs3/pubsys2/xml/en/procedures/122/122-t05-1622qsk50-60.html)
Procedure Title	Injector Solenoid Driver Cylinder 16 Circuit - Current Below Normal or Open Circuit
Update Reason	Revise Step 4A, Yes Repair and Next Step to Escalate or call for assistance.

Date	21-Oct-2019
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	105-019-032 (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)
Procedure Title	Engine Control Module Calibration Code
Update Reason	Added the following statement to the Preparatory Steps of the procedure: "Verify vehicle odometer is equal to the value recorded before ECM removal. Contact OEM service location if values are incorrect."
Date	11-Dec-2019
Section	Section 19 - Electronic Controls
Procedure Number	122-019-427 (/qs3/pubsys2/xml/en/procedures/122/122-019-427.html)
Procedure Title	Engine Control Module ROM Boot
Update Reason	Updated part numbers to reflect with Service Tool Bulletin, 3377791-26.
Date	16-Apr-2020
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5119 (/qs3/pubsys2/xml/en/procedures/122/122-fc5119.html)
Procedure Title	Fuel Pump Oil Pressure Sensor Circuit - Voltage Above Normal or Shorted to High Source
Update Reason	Added fault code to manual.
Date	16-Apr-2020
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc5121 (/qs3/pubsys2/xml/en/procedures/122/122-fc5121.html)
Procedure Title	Fuel Pump Oil Pressure Sensor Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	Added fault code to manual.

Date	16-Apr-2020
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc7313 (/qs3/pubsys2/xml/en/procedures/122/122-fc7313.html)
Procedure Title	Engine Fuel Pump Oil Pressure - Data Valid But Below Normal Operating Range - Most Severe Level
Update Reason	Added fault code to manual.
Date	16-Apr-2020
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-fc7314 (/qs3/pubsys2/xml/en/procedures/122/122-fc7314.html)
Procedure Title	Engine Fuel Pump Oil Pressure - Data Valid But Below Normal Operating Range - Moderately Severe Level
Update Reason	Added fault code to manual.
Date	14-May-2020
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	56-019-679 (/qs3/pubsys2/xml/en/procedures/56/56-019-679.html)
Procedure Title	Fuel Pump Lubricating Oil Supply Pressure Sensor
Update Reason	Added procedure to manual.
Date	21-May-2020
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-5119 (/qs3/pubsys2/xml/en/procedures/122/122-t05-5119.html)
Procedure Title	Fuel Pump Lubricating Oil Pressure Sensor Circuit- Voltage Above Normal or Shorted to High Source
Update Reason	Added fault code troubleshooting tree to manual.

Date	21-May-2020
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-5121 (/qs3/pubsys2/xml/en/procedures/122/122-t05-5121.html)
Procedure Title	Fuel Pump Lubricating Oil Pressure Sensor Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	Added fault code troubleshooting tree to manual.

Date	21-May-2020
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-7313 (/qs3/pubsys2/xml/en/procedures/122/122-t05-7313.html)
Procedure Title	Engine Fuel Pump Oil Pressure - Data Valid But Below Normal Operating Range - Most Severe Level
Update Reason	Added fault code troubleshooting tree to manual.

Date	21-May-2020
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	122-t05-7314 (/qs3/pubsys2/xml/en/procedures/122/122-t05-7314.html)
Procedure Title	Engine Fuel Pump Oil Pressure - Data Valid But Below Normal Operating Range- Moderately Severe Level
Update Reason	Added fault code troubleshooting tree to manual.

Date	17-Jul-2020
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	122-019-004 (/qs3/pubsys2/xml/en/procedures/122/122-019-004.html)
Procedure Title	Barometric Pressure Sensor
Update Reason	Reorganized procedure to include information pertaining to both mechanical and electronic actuated injectors.

Date	21-Jul-2021
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	122-019-450 (/qs3/pubsys2/xml/en/procedures/122/122-019-450.html)
Procedure Title	Exhaust Gas Temperature Sensor Converter
Update Reason	Added procedure to manual.

Last Modified: 07-Dec-2021
